

The Safety Substrate Underneath Clinical AI

Deterministic verification infrastructure for hospital systems deploying AI-assisted clinical decision support. Published methodology. Provisional patents filed.

The Problem

Clinical AI systems are being deployed faster than the safety infrastructure underneath them. Most clinical AI wraps a language model in a policy layer and calls it safe. When the reasoning is wrong, the wrapper is also wrong — because they share the same process. There is no independent verification that the AI's output is safe for this patient, right now, given everything the system knows.

What PharmMesh Does

PharmMesh builds a deterministic verification layer that sits between the AI's output and the patient. The verification layer runs independently of the AI reasoning — neither can override the other. Every clinical claim the AI produces is checked against deterministic rules before it reaches a clinician. Every case produces a structured audit record designed to survive regulatory inspection.

Independent Verification	Cross-Silo Aggregation	Audit Trail
The safety layer cannot be disabled by the AI reasoning above it. When the AI is wrong, the verifier catches it before the patient.	Designed to aggregate medication data across EHR, cross-pharmacy, and monitoring program sources— closing the gap between what one institution sees and what the patient is actually taking.	Every gate firing, every input, every override produces a structured record meeting the audit expectations of OMB M-25-21 for federal high-impact AI system

The Escape Register

Every time the system encounters a new way clinical AI can fail, the failure is documented, a permanent code-level safeguard is built, and the safeguard is validated to never regress. The safety net reshapes itself based on what falls through. Each near-miss at one institution can be broadcast as a deterministic rule to every connected institution — without protected health information crossing the boundary. The institutions get safer together.

What PharmMesh Is Not

PharmMesh does not replace clinical judgment. It does not generate diagnoses. It does not make treatment decisions. It provides deterministic verification that makes other people's AI defensible.

Why Now

HTI-1 §170.315(b)(11) — 31 transparency attributes required for every predictive clinical decision support intervention in certified health IT. In effect.

OMB M-25-21 — Audit trail requirements for high-impact AI in federal programs. In effect.

CMS-0057-F — FHIR-based prior authorization API requirements activating across 2026–2027.

CMS POLY-ACH — Concurrent anticholinergic medication use becomes a Medicare Part D Star Rating measure with Star Year 2027 financial impact.

Current Status

Methodology	Published open-access preprint on Research Square (CC-BY 4.0). Peer review in progress.
Intellectual Property	Provisional patent corpus filed covering the safety architecture, escape register methodology, and federated rule broadcast.
Validation	Validated across multiple test scenarios with zero safety regressions across three consecutive infrastructure changes.
Stage	Pilot-ready. Seeking design partners for first clinical deployment